Linux Network Programming: Raw Sockets & PF_PACKET

Coverage: Data Link Provider Interface (DLPI), BSD Packet Filter (BPF), Linux PF_PACKET / AF_PACKET (SOCK_RAW, ETH_P_ALL), Promiscuous Sniffer

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1. Linux PF_PACKET Raw Sockets

- AF_PACKET domain bypasses kernel TCP/IP stack to capture/inject raw L2 Ethernet frames.
- Promiscuous Mode (IFF_PROMISC): Disables MAC filtering to capture all link traffic.

■ KTU University High-Yield Exam Questions

- Q: Explain the architecture of PF_PACKET raw sockets and write a packet sniffer in C. (8 Marks)